

LECTURE L25-26 • MODULE 4 • CO4

Feature engineering and selection

Week 13 | Slides: DATA209_Module_4_Feature_Engineering_and_Structure.pptx, pages 12-20 | Laboratory: P25-26

TOPICS COVERED

- Feature construction; interaction and polynomial features
- Curse of dimensionality
- Feature selection: filter, wrapper, embedded

A feature is a question

Feature engineering is where domain knowledge enters the pipeline. It has the highest return on effort of any step and is the one most often skipped.

Six ways to build one:

- **Ratios** — BMI from weight and height; conversion rate from clicks and views.
- **Differences** — days since last visit; deviation from a group mean.
- **Aggregates** — per-entity mean, max or count.
- **Date parts** — month, weekday, is-weekend, hour. One date becomes several features.
- **Binning** — age into clinical bands. Loses information, gains robustness.
- **Indicators** — `was_missing`, is-zero, exceeds-threshold. Cheap, interpretable, often strong.

Every constructed feature needs a one-line rationale. If you cannot say why it should matter, do not build it.

Interactions and polynomials

Interaction features are products of two variables, capturing effects neither shows alone. Linear models cannot discover these; trees can, partially. **Polynomial features** are powers of a variable, letting a linear model fit a curve.

`PolynomialFeatures(degree=2)` on 8 columns yields 44 — count before you generate. Both overfit quickly; pair with regularisation and validation.

Generate deliberately, not exhaustively. Automatic expansion is the fastest route into the curse of dimensionality.

Three families of selection

- **Filter** — score each feature independently of any model (correlation, chi-square, mutual information, variance). Fast, model-agnostic, blind to interactions.

- **Wrapper** — search subsets by training a model on each (RFE, forward/backward selection).

Accurate for that model, expensive, overfits easily.

- **Embedded** — selection happens during fitting: Lasso shrinks coefficients to zero, trees report

importances. A good default balance.

	FILTER	WRAPPER	EMBEDDED
Cost	very low	high	moderate
Model-specific	no	yes	yes
Sees interactions	no	yes	partly
Overfitting risk	low	high	moderate

Run a filter first to remove the obviously useless, then an embedded method on what survives. Reserve wrappers for small feature sets.

Agreement is the signal

Applying five methods to the shoppers feature set, `PageValues`, `has_page_value` and `ExitRates` are chosen by all five; several are chosen by only one. **A feature selected by one method alone deserves scrutiny.**

The honest test of feature engineering is whether the constructed features survive selection — and whether a smaller set performs comparably. On the shoppers data, 18 features give cross-validated ROC-AUC 0.906 and a 6-feature consensus set gives 0.875. The smaller set is cheaper, more stable and easier to explain; whether the difference is worth it is a judgement you must state.

Select **inside** cross-validation. Selecting on the full dataset and then validating is leakage.

KEY TERMS

Feature construction	Interaction feature	Polynomial feature	Binning	Filter method
Wrapper method	Embedded method	RFE	Mutual information	Lasso

COMMON ERRORS TO AVOID

- Generating polynomial features exhaustively without counting them
- Selecting features on the full dataset before splitting
- Trusting a single selection method's ranking
- Building features with no stated rationale

READINGS — ALL FREE TO ACCESS

1. scikit-learn — feature selection
https://scikit-learn.org/stable/modules/feature_selection.html
2. Kuhn & Johnson — Feature Engineering and Selection
<http://www.feat.engineering/>
3. Kaggle Learn — Feature Engineering
<https://www.kaggle.com/learn/feature-engineering>

TEST YOURSELF

1. Give three features you could derive from a single timestamp column.
2. Why can a filter method miss a genuinely useful feature?
3. You have 40 features and 300 rows. Which selection family, and why?
4. Why must feature selection sit inside cross-validation?